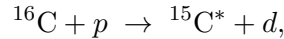


A Student's Guide to the $^{16}\text{C}(p, d)^{15}\text{C}$ Analysis

(a1975, AT-TPC active target, genfit reconstruction)

1 What you are measuring

The a1975 experiment is a ^{16}C beam (~ 192 MeV, ~ 12 MeV/u) stopped in an **H₂ gas active target** at 1 bar. The gas is *both* the proton target and the tracking medium. You will analyze the **neutron pickup** reaction



i.e. $^{16}\text{C}(p, d)^{15}\text{C}$, by detecting the outgoing **deuteron**. From the deuteron's kinetic energy T_d and lab angle θ_{lab} you reconstruct the ^{15}C excitation energy E_x via two-body relativistic kinematics, and you study which ^{15}C states are populated and how strongly (the angular distribution).

Goal of this guide: starting from the reconstructed data, produce a clean ^{15}C excitation-energy spectrum and its angular dependence, and understand the two systematics that matter: *triton contamination* and the E_x -vs- θ_{cm} drift.

2 Setup

All commands below are run from the (p,d) macro directory after sourcing the environment:

```
cd ~/fair_install/ATTPCR00Tv2-OpenKF/macro/Unpack_HDF5/a1975/UKF
source ~/fair_install/ATTPCR00Tv2-OpenKF/build/config.sh
```

The raw data and reconstructed files live on the F: drive (mount it in WSL with `sudo mount -t drvfs F: /mnt/f` if needed):

- /mnt/f/a1975/h5/ raw HDF5 (one file per run, run_0106–run_0189)
- /mnt/f/a1975/reco/ reconstructed *_reco.root (hits + pattern/tracks) + *.FRIB.root (ion chamber)
- /mnt/f/a1975/reco_pd/ your (p,d) fit outputs go here

The analysis macros are in `pp/`; the interactive ones need an X display (they work over WSLg; the static ones write a PNG you open with `explorer.exe`).

3 The pipeline at a glance

raw HDF5 → [unpack + PSA + space-charge + PRA] → *_reco.root → [deuteron genfit fit] →
*_genfitter_pd.root → [cache] → pd_kin.root → [plots]

Key idea: pattern recognition (PRA, finding the tracks) does not depend on what particle you assume, so the *_reco.root files are shared by all channels. For (p,d) you only need to *fit* the existing tracks with the **deuteron** hypothesis. You do *not* re-run the reconstruction.

4 Step 1 — Fit the deuterons

fitGenfitter_a1975.C reads a *_reco.root, keeps only tracks inside the deuteron PID band (pid/deuteron_band.json), fits each as a deuteron with genfit, and writes the fitted kinematics. Its defaults are already the deuteron hypothesis, so for one run:

```
root -l -b -q 'pipeline/fitGenfitter_a1975.C("run_0106", -1, "/mnt/f/a1975/reco/", "_pd", "/mnt/f/a1975/reco_pd/")'
```

The arguments are (run, nEvents=-1 for all, inputDir, outputSuffix, outputDir). The output is /mnt/f/a1975/reco_pd/run_0106_genfitter_pd.root. Run the whole experiment on 4 cores:

```
ls /mnt/f/a1975/reco/run_*_reco.root | grep -oE 'run_[0-9]+' | sort -u > /tmp/pd_runs.txt
cat /tmp/pd_runs.txt | xargs -P4 -I{} \
  root -l -b -q "pipeline/fitGenfitter_a1975.C(\"{}\", -1, \"/mnt/f/a1975/reco/\", \"_pd\", \"/mnt/f/a1975/reco_pd/\")"
```

A deuteron is rare ($\sim 1.5\%$ of tracks), so most events show 0/1 **tracks fitted** — that is the PID gate doing its job, not an error. Expect $\sim 30,000$ deuterons over the 84 runs.

5 Step 2 — Cache the kinematics

The fit files are large; cache just the per-deuteron numbers you need into a flat ntuple. The cache also pulls the **ion-chamber (IC)** value from the *_FRIB.root files (the IC tags good ^{16}C beam particles).

```
cat /tmp/pd_runs.txt | xargs -P4 -I{} root -b -q 'pp/cache_pd_run.C("{}","/tmp/pdkin_{}.root")'
hadd -f /tmp/pd_kin.root /tmp/pdkin_run_01*.root
```

This makes /tmp/pd_kin.root, an ntuple pk with branches ke, theta, vz, chi2ndf, ic, run. All later plots read this single file (fast).

6 Step 3 — The excitation-energy spectrum

From T_d and θ_{lab} the macros compute $E_x(^{15}\text{C})$ with relativistic two-body kinematics. Make the 4-panel overview:

```
root -l -b -q 'pp/plot_pd.C' # -> /tmp/pd_spectrum.png
explorer.exe "$(wslpath -w /tmp/pd_spectrum.png)"
```

You should see (i) the E_x spectrum with a sharp ^{15}C ground-state peak near 0 and excited states ($\sim 3.10, 4.66$ MeV); (ii) the deuteron T_d -vs- θ_{lab} band lying on the analytic kinematic line; (iii) E_x vs θ_{cm} ; (iv) vertex- z vs E_x .

Cuts. plot_pd.C applies a χ^2/ndf cut and the IC beam gate (950–1350) by default. These remove badly-fit and non-beam events.

7 Step 4 — Angular dependence

A discrete state should appear at the same E_x at every angle, and its population (cross section) changes with angle. Plot E_x in θ_{cm} slices:

```
root -l -b -q 'pp/plot_pd_angbins.C(10,5,12)'          # start 10 deg, 5 deg wide, 12
               bins
explorer.exe "$(wslpath -w /tmp/pd_angbins.png)"
```

Arguments are (thcmStart, thcmWidth, nbins, ...). The ground-state resolution is best at forward angles ($\sigma \approx 0.15$ MeV at 20–30°) and degrades toward back angles.

8 Step 5 — Correct the E_x -vs- θ_{cm} drift (important!)

The reconstructed ground state is *not* flat in θ_{cm} : its centroid drifts by ~ 0.9 MeV across the angular range. This is a kinematic/reconstruction systematic, and it **smears the integrated spectrum**. The fix measures the g.s. centroid versus θ_{cm} and subtracts it event-by-event:

```
root -l -b -q 'pp/excorr_pd.C'                          # -> /tmp/pd_excorr.png
```

The left panel shows the measured drift (and the curve subtracted); the right panel shows the integrated ^{15}C spectrum before and after. The ground-state FWHM improves from ~ 1.7 MeV to ~ 0.43 MeV. *This is the single most important correction for the integrated spectrum*. The same correction is built into `plot_pd_angbins.C` (argument `driftCorr`, on by default) and into the interactive explorer (a checkbox).

9 Step 6 — Beware triton contamination

$^{16}\text{C}(p,t)^{14}\text{C}$ produces **tritons**. The deuteron PID gate is loose enough to admit some of them, and genfit fits a triton *as a deuteron* with a perfectly good χ^2 (so you cannot remove tritons with a χ^2 or IC cut — we checked). They show up as a faint second band above the deuteron locus in T_d vs θ_{lab} . Look at the PID plane:

```
root -l -b -q 'pp/pid_pd.C(106,113)'                    # sqrt(dE/dx) vs Brho + the gate
explorer.exe "$(wslpath -w /tmp/pid_pd.png)"
```

The proton, deuteron and triton bands are separated in the $\sqrt{dE/dx}$ - $B\rho$ plane; the red polygon is the current (loose) deuteron gate. **The fix is to tighten the gate** so the triton band is excluded — because the species separation must come from the gate, not from the fit. Use the interactive editor (draw a polygon, save it to JSON, then re-run Step 1 with the new gate):

```
root -l 'pp/pid_gate_edit.C'                            # Draw new gate -> click vertices, double-click to
               close -> Save JSON
```

10 Step 7 — Interactive exploration

For tuning cuts and binning live (needs a display):

```
root -l 'pp/explore_pd.C'
```

Controls: beam energy, χ^2/ndf , θ window, IC window, **vertex- z bounds**, E_x binning, plus checkboxes for the **drift correction** and the ^{15}C reference lines. Adjust and press **Redraw**; **Save PNG** to export.

11 Physics checklist

- The ^{15}C ground state ($1/2^+$) should sit at $E_x = 0$ after calibration; the first excited state is at 0.740 MeV ($5/2^+$, on the g.s. shoulder).
- Prominent strength near 3.10 and 4.66 MeV; broad structure at 6–8 MeV.
- The deuteron $T_d(\theta)$ band must follow the analytic ground-state kinematic line — if it does not, suspect a wrong beam energy, B -field sign, or frame convention.

12 Troubleshooting

- “*Macro is slow / hangs.*” The PID macros only need the **AtPIDEvent** branch; they already disable the big track branch. The IC gate reads the large FRIB files and is therefore slower — it is opt-in in **pid_pd.C**.
- “*/tmp/pd_kin.root missing.*” **/tmp** clears on reboot — rebuild the cache (Step 2).
- “*Gate file not found.*” Run from the **UKF** directory so **pid/...json** resolves.
- “*Interactive window does not open.*” The explorer/editor need **X** (WSLg); the **-b** flag means batch — drop it for the GUIs.